The Thermodynamics of Generative AI – Building a Diffusion Model from Scratch

Week 1

Leonard Heinze

Salmeri Elisa

Paul Jachmann

Daniel Rodau

Ian Schönherr

Valentyn Skalenchuk

Simulating the Forward Process

Theoretical Background - Stochastic Differential Equation

$$dx_t = a(x_t, t)dt + b(x_t, t)dWt$$

x_t … The value of the system at a time t

$a(x_t, t)dt$ … Drift term

$b(x_t, t)dW_t$ … Diffusion term

dW_t … The Wiener Process

Langevin Dynamics

$$m d\mathbf{v} = -\gamma v dt - \nabla U(\mathbf{x_t}) dt + \sqrt{2\gamma k_B T} dW_t$$

m

γ … mass of the particla

... friction

$\nabla U(\mathbf{x})$

... gradient of the potential

$\sqrt{2\gamma k_B T} dW_t$

... random thermal noise

Overdamped Langevin equation

Overdamped Limit → viscous forces are so big, that the particles momentum dissipates almost instantly:

$$dx_t = -\frac{1}{\gamma}\,\nabla U(x_t)dt + \sqrt{2D}\,dW_t$$

$D = \frac{k_B T}{\gamma}$ … Diffusion coefficient

Forward Diffusion

use overdamped Langevin eq:

$$dx_t = -\nabla U(x_t)dt + \sqrt{2D}dW_t$$

$$-\frac{1}{2}\beta(t)x_t$$

$$\sqrt{\beta(t)}$$

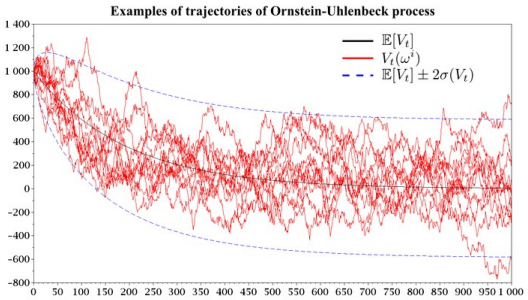
$\sqrt{\beta(t)}$:noise schedule

Obtain the **Forward Diffusion Process:**

$$dx_t = -\frac{1}{2}\beta(t)x_t dt + \sqrt{\beta(t)}dW_t$$

The Ornstein-Uhlenbeck Process (OU)

– mean-reverting Brownian motion



$$dx_t = -kx_t dt + \sqrt{2D} dW_t$$

Restoring Part

Driving Part/Diffusion part

→ Over an infinite number of time steps, the system converges to a stable Gaussian distribution.

Forward Diffusion

– make the image noisy, each steps t (time steps) add noise

→ reverse process: generate the image

– markov chain x_t: x_0 ….. X_1000

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t, \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

$$x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Signal Noise

https://medium.com/@kemalpiro/step-by-step-visual-introduction-to-diffusion-models-235942d2f15c